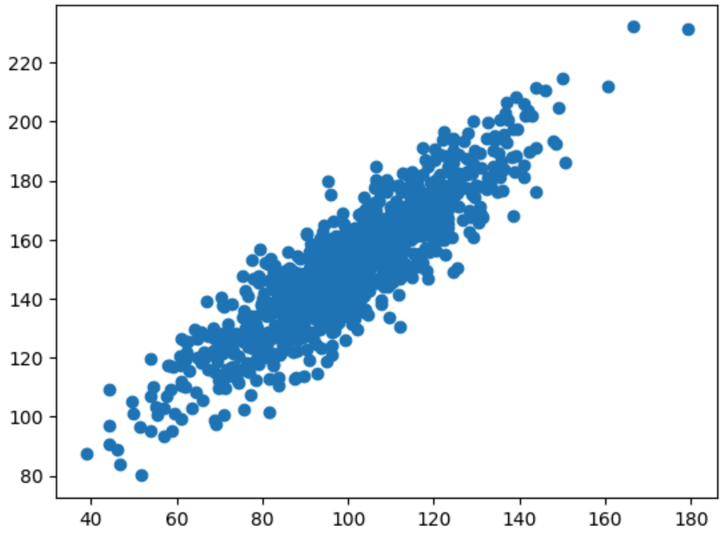
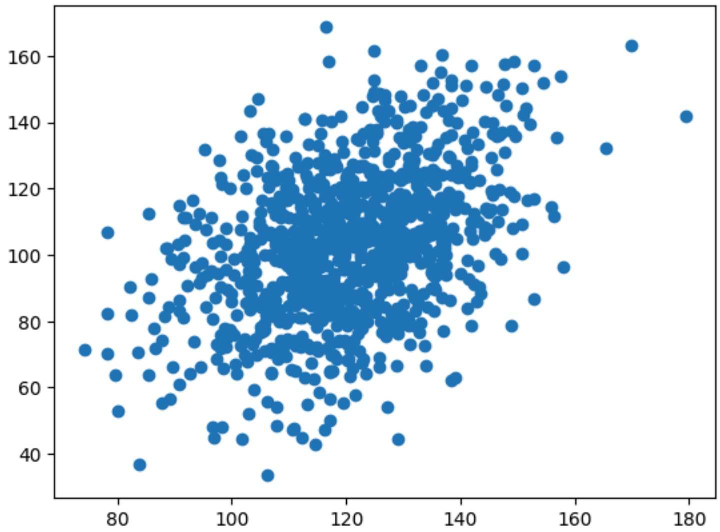
```
# calculate the Pearson's correlation between two variables
from numpy import mean
from numpy import std
from numpy import cov
from numpy.random import randn
from numpy.random import seed
from matplotlib import pyplot as plt
import seaborn as sns

from scipy.stats import pearsonr
# seed random number generator
seed(1)

# prepare data
data1 = 20 * randn(1000) + 100
data2 = data1 + (10 * randn(1000) + 50)

# calculate covariance matrix
covariance = cov(data1, data2)

# calculate Pearson's correlation
corr, _ = pearsonr(data1, data2)

# plot
plt.scatter(data1, data2)
plt.show()

# summarize
print('data1: mean=%.3f stdv=%.3f' % (mean(data1), std(data1)))
print('data2: mean=%.3f stdv=%.3f' % (mean(data2), std(data2)))
print('Covariance: %.3f' % covariance[0][1])
print('Pearsons correlation: %.3f' % corr)
```




```
data1: mean=100.776 stdv=19.620
data2: mean=151.050 stdv=22.358
Covariance: 389.755
Pearsons correlation: 0.888
```

```
from scipy.stats import pearsonr
# seed random number generator
seed(1)

# prepare data
data1 = 15 * randn(1000) + 120
data2 = 0.6 * data1 + (20 * randn(1000) + 30)

# calculate covariance matrix
covariance = cov(data1, data2)

# calculate Pearson's correlation
corr, _ = pearsonr(data1, data2)

# plot
plt.scatter(data1, data2)
plt.show()
```

```
# summarize
print('data1: mean=%.3f stdv=%.3f' % (mean(data1), std(data1)))
print('data2: mean=%.3f stdv=%.3f' % (mean(data2), std(data2)))
print('Covariance: %.3f' % covariance[0][1])
print('Pearsons correlation: %.3f' % corr)
```




```
data1: mean=120.582 stdv=14.715
data2: mean=102.896 stdv=22.590
Covariance: 136.682
Pearsons correlation: 0.411
```

Conclusion

I modified the dataset by reducing the dependency between the two variables and introducing additional random variation. After rerunning the analysis, the covariance remained positive (136.682), indicating that the variables still tended to increase together. However, the Pearson correlation coefficient decreased to 0.411, which indicates a moderate positive linear relationship. Compare with the original dataset, the increased varaiability in data2 weakened the strength of the correlation. This exercise demonstrates that covariance identifies the direction of the relationship, whilst Pearsons correlation measures both the direction and the strength of the linear association between variables.

```
Start coding or generate with AI.
```